

Comparative Study of Genetic Algorithm and Ant Colony Optimization Algorithm Performances for Robot Path Planning in Global Static Environments of Different Complexities

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Abstract—This paper presents the application of Genetic Algorithm (GA) and Ant Colony Optimization (ACO) Algorithm for robot path planning (RPP) in global static environment. Both algorithms were applied within global maps that consist of different number of free space nodes. These nodes generally represent the free space extracted from the robot map. Performances between both algorithms were compared and evaluated in terms of speed and number of iterations that each algorithm takes to find an optimal path within several selected environments. The effectiveness and efficiency of both algorithms were tested using a simulation approach. Comparison of the performances and parameter settings, advantages and limitations of both algorithms presented herewith can be used to further expand the optimization algorithm in RPP research area.

I. INTRODUCTION

PATH planning is an important task in autonomous mobile robot to enable the robot navigation system to identify a safe path (without colliding with obstacles) to goal. Path planning research covers a wide area of robotics research that includes path planning in static [1, 2] and dynamic environments [3, 4]. In static environment, the environment is assumed to be known and the global path can be generated in advanced based on the provided map. This is known as a model-based approach which is implemented offline. In a dynamic environment, the mobile robot will respond to unexpected situation and the map is updated to create a new path in responds to the environment change. This is known as sensor based or online approach where the new path is constructed once the sensor detects that there are changes in the environment. This research is focused on path planning in static environment.

Generally, there are two main elements involved in the process of global path planning which is the mapping and the path planning algorithms (PPAs). The environment will be mapped in the initial process. The existence of these maps will simplify the process of finding the optimal path when the PPAs are applied to the maps. Example of maps created and used by previous researchers for their RPP purposes is

C-Space such as cell decomposition; grid map [2, 5, 6], graph; visibility [7] and MAKLINK graph [3]. Within these maps, the area of feasible (no obstacles) and non feasible nodes (obstacles) have been configured before PPAs are applied to find the paths. The selection and utilization of maps in a research must be compatible with the selected PPA and this depends on the objectives and applications of the research itself.

The evolution of PPAs shows improvements of PPA from one generation to the next generation. The later generation of PPAs have been created to be more adaptive and able to work within the robot environment itself [8]. Several PPA traditional approaches have been developed by previous researchers such as artificial potential fields [9], neural networks [10], heuristic algorithms [1, 6] and etc. Each method has its own strength and limitations over others in certain aspects of path planning itself. The evolution of traditional approaches produced a widely used heuristic algorithm known as A* algorithm [1]. This algorithm is widely used not only in static environments but it subsequently evolved to D* algorithm [6] which is then widely used in dynamic environments.

However, since appearances of artificial intelligences [11], the problem of path planning has been view as one of an optimization problem. The idea of utilizing artificial technologies in developing intelligent agent for RPP purposes has grown since then. Currently, researchers use neural networks [10], evolutionary computation [12, 13] and swarm intelligence [14] in RPP research. This newer approach enables the identification of optimal robot path that satisfies optimization criteria such as shortest and fast with small computational time. The wide application of PPAs such as Genetic Algorithms (GA) [2, 15] and Ant Colony Optimization (ACO) algorithm [3, 4, 16] in RPP research is because it can produce the optimal path effectively compared to the traditional approaches.

The research presented herewith investigates the effectiveness of GA and ACO applied in different complexity of free space nodes in a global map. The performances of the algorithms were evaluated in terms of speed and number of iterations when both algorithms were applied in different complexity of free space nodes in global maps. The global maps that were constructed consists of different number of obstacles, different location of obstacles, or different type of mapping algorithms used which produced different complexity of nodes. It is hoped that findings from this research can be used by researchers in the RPP optimization area.

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This paper discusses GA and ACO algorithm construction including mapping of simple environments and complex environments that were used in this case study. Then the comparison of performances (using simulated data) between GA and ACO in terms of speed and number of iterations in different maps is presented. The pattern of the parameter settings for both algorithms in different complexity of environment as identified is also included. Finally the discussion of the usefulness of these algorithms in RPP research was discussed.

II. RESEARCH BACKGROUND

This section covers the utilization of GA and ACO in RPP research areas where different cases have different methods of presenting the solution to find an optimal robot path.

A. Genetic Algorithm

Since its appearance 1975 [12, 13], GA has been used in solving many robot path planning optimization problems. GA is a search techniques inspired by evolutionary biology where it work based on principle of the fittest of the chromosomes. With its ability to work with parallel search techniques, the use of GA contributed to the success in many robot path planning research. For example, Gihan Nagib et al [5] proposed the use of GA to find robot path based on map of free space nodes. Kuzuo Sugihara [15], R. Ramakrishnan [17] also proposed the used of GA with different encoding techniques to ensure GA can find optimal path without depending on the feasible nodes given in the map. Yangrong Hu [18] modified classical GA by incorporating the domain knowledge into specialized operator to improve GA performances when it works in environments that consists of obstacles. From the above literature study, it can be concluded that GA performances depends on the way the solution is encoded in chromosomes, accuracy of the fitness function and variation of genetic operators which determine the whole process of GA.

B. Ant Colony Optimization Algorithm

ACO, compared to GA is a newer optimization method. Introduced by Marco Dorigo [14] in approximately 1992, the application of this algorithm in RPP research increased rapidly as it is a powerful tool for solving hard combinatorial optimizations problem. ACO was inspired by analogy of behavior of real ants, when looking for foods. Tan Guan Zheng [3] proposed the use of ACO to find robot path based on map of MAKLINK graph. Hao Mei [4] combined ACO with Artificial Potential Field to produced the path planning in dynamic environment. Gengqian et al [16] have proven that ACO can find optimal path in their grid map by proposing its own probability equation. However, a literature study shows that the application of ACO to solve RPP problems has not been explored in detail.

III. RESEARCH METHODOLOGY

Fig. 1 below illustrates the proposed method applied within this research. At the beginning stage, the robot environment needs to be mapped using an appropriate global map as discussed in section A below. This map will create

an output of nodes represented by xy coordinates. Then, GA and ACO will start to initialize the population of path using its own approaches from start to goal by using all the provided nodes including the start, goal and all intermediate nodes. During the initialization, the integer number represents by each node will be used. However, during the evaluation, the real xy coordinates will be used. At the end of the process, the optimal path will be found.

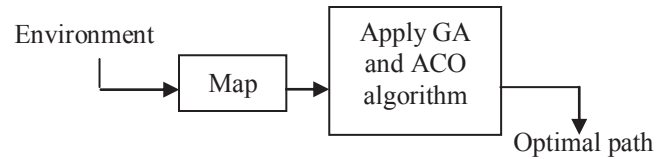


Fig.1. Proposed Method

B. Environment Modeling

In this research, a 2D grid map as shown in Fig. 2 below was used. The free space nodes (white grid) represent the area the robot can traverse including the robot size. The obstacles area (black grid) represents the boundary of obstacles with the safety region and the yellow grid represents the feasible free space nodes that can be traversed by the robot. However, during construction of the map used for this study, the obstacles was assumed to have been eliminated and only a route of feasible nodes is left as shown in Figure 4,5 and 6.

By assuming the 10 X 10 cm grid map is the size of the map, three different complexities of free space nodes have been developed. A simple environment consists of 12 numbers of nodes (Fig. 4), average with 22 numbers of nodes (Fig. 5) and complex environment have 63 numbers of nodes (Fig. 6). These feasible free space nodes have been located randomly within the grid map for this research comparative study purposes. By using this maps, the algorithm will asses all feasible paths using the available feasible nodes as shown in population of path depicted in Fig. 3 below. However during implementation, there will be non feasible nodes produced during the generation of the optimal path.

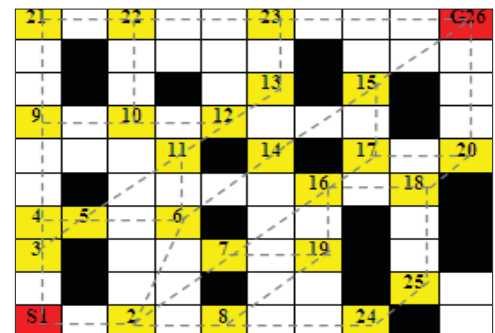


Fig. 2. Global feasible map with obstacles

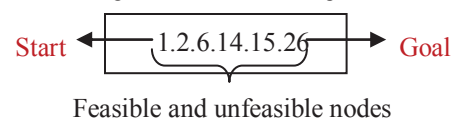


Fig. 3. A sample of path population consists of feasible nodes of Fig. 2

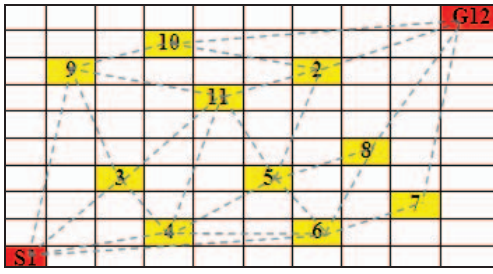


Fig. 4. Simple free space map

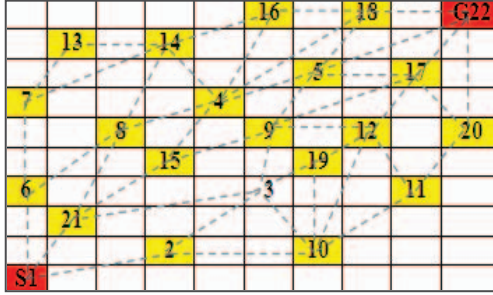


Fig. 5. Average free space map

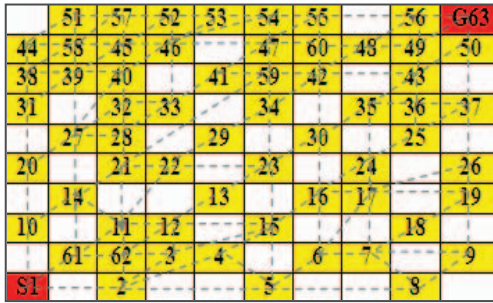


Fig. 6. Complex free space map

B. Genetic Algorithm Design for RPP

The outline of GA is given in Fig. 7. The initial solutions of the RPP problem will initialize in population randomly. In the first case, the population will initialize based on the feasible nodes provided in the global map only. With the complete population, the fitness is evaluated by using the formula below:

$$\text{Fitness node} = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2} \quad (1)$$

$$\text{Total Fitness} = \begin{cases} \sum \text{Fitness node} & ; \text{Feasible} \\ 100 & ; \text{Unfeasible} \end{cases} \quad (2)$$

After the fitness of each population has been evaluated, it will be ranked using an elitism approach. The shorter path will be represented with a high fitness value and will be selected to be carried forward to the next generation while the long path represented with a low fitness value will be eliminated and removed from the population. The good parents which is carried forward to the next generation will produce the diversity of population that consists of a good child from the genetic operators process. Then this process is repeated until all of the GA population found the same

optimal path with no difference of the fitness value where the distance is equal to 0. It is at this moment, that the solution converges. However, type of GA and important parameters specifications related with GA used in this experimental research is defined in Table I below:

TABLE I
GA PARAMETER SPECIFICATIONS

GA Properties	Parameter
Type of GA	Classical GA
Chromosomes type	Fixed length chromosomes
Population size	Varies, depends on cases
Chromosomes length	Varies, depends on cases
Selection type	Elitism
Crossover type	Two point crossover
Mutation type	Flip bit
Crossover rate	0.75
Mutation rate	0.75
Convergence criteria	Cmax-Cmin≤0.00001

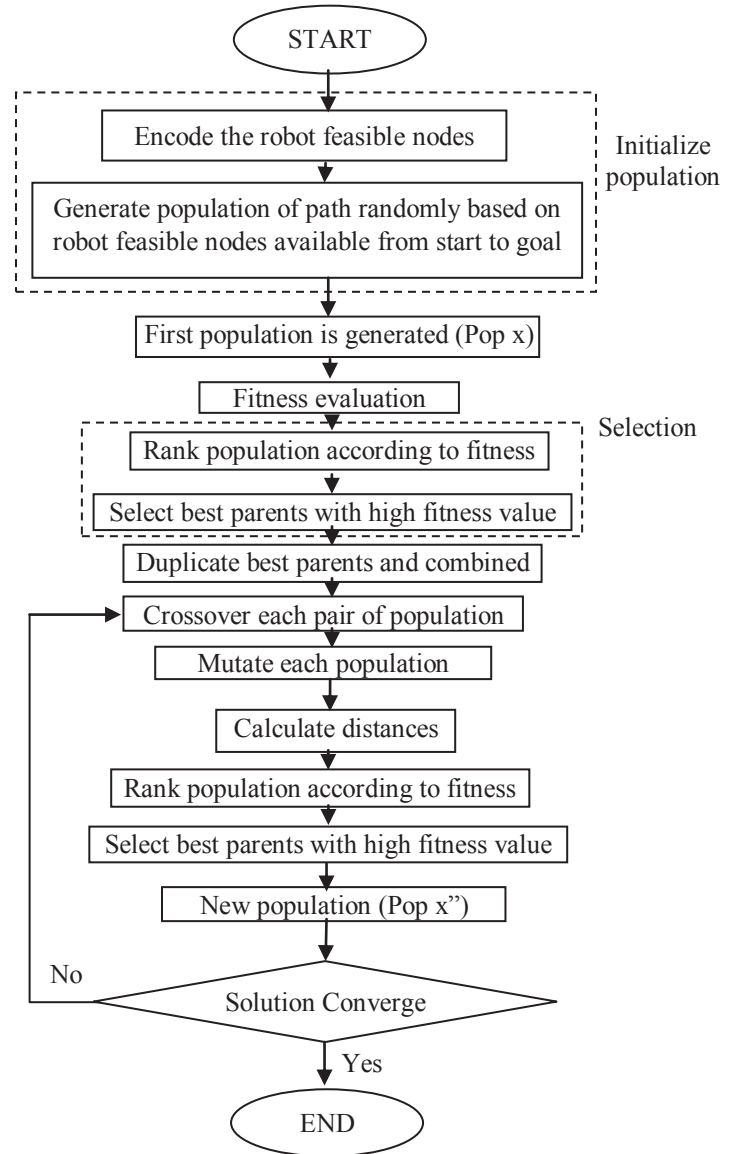


Fig. 7. Outline of GA for RPP of a mobile robot

C. Ant Colony Optimization Design for RPP

The ACO algorithm used in this experiment is the Ant System (AS) algorithm proposed by Marco Dorigo [14]. However, equation (3), a new heuristic equation of the state transition rules, which is more suitable with applications of this research, was used. The evaluation fitness and ACO parameter setting was created based on the requirements of this research.

The design of AS for robot path planning was divided into three important rules which are state transition rules, local update rules and global update rules. At the beginning, ants will determine the next node to be visited by using the state transition rules based on heuristic and pheromone laid down by the ants as shown in derivation below:

$$\text{Probability } ij = \text{heuristic} * \text{pheromone} \quad (3)$$

$$= [(1/\text{distance between vector start to subpath and start to perpendicular subpath with reference goal})^\beta * (\text{trail}/\sum \text{trail})^\alpha]$$

* β =heuristic coefficient, α =pheromone trail coefficient

An accurate value of distance by heuristic equation and the higher amount of pheromone of the visited node will be obtained by the ants that have higher probability to choose that nodes. Within these rules, ants can balance between the exploration and exploitation from the relatives' coefficient provided, known as alpha and beta. During the construction of the path, the pheromone will be reduced locally by the given evaporation rate by using the formula of update local rules below:

$$T_{ij}(\text{new trail}) \leftarrow (1-\rho) * T_{ij}(\text{old trail}), \quad (4)$$

* ρ =evaporation rate

After all the ants complete the path to goal, then the process of global updating is applied where ants will deposits its pheromone based on the path distance.

$$t_{ij} \leftarrow t_{ij} + \sum \Delta t_{ij}^k \quad (5)$$

Δt_{ij}^k = amount pheromone of ant m deposits on the path it has visited. It's defined as below:

$$\Delta t_{ij}^k = \begin{cases} Q/C^k & ; \text{if arc } (i,j) \text{ belongs to path } P^k \\ 0 & ; \text{otherwise} \end{cases} \quad (6)$$

where Q is number of nodes and C^k is the length of path P^k built by the ants

The amount of pheromone will continuously be updated until it attracts more ants from the next generation to follow the shorter path. Finally, the optimal robot path is found by using behavior of ants' concept as shown in Fig. 8 below.

The parameter specifications of ACO utilized in this experiment is shown in Table II below.

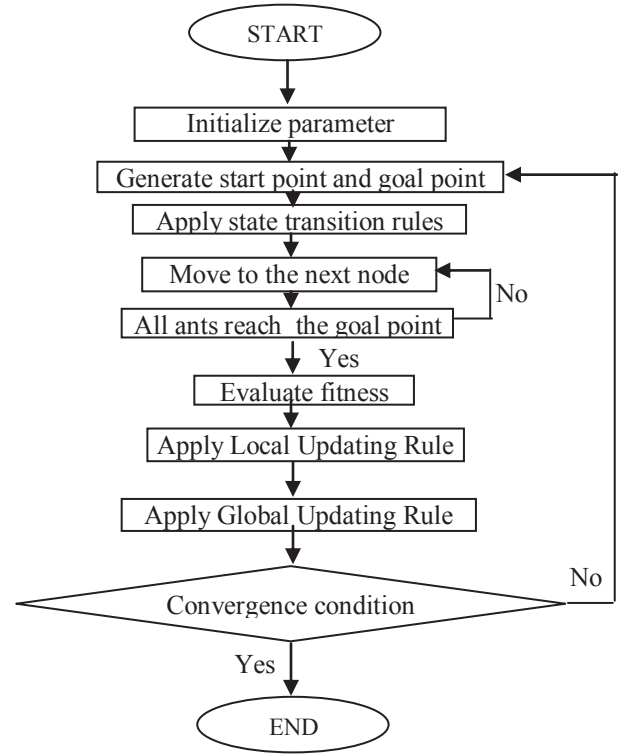


Fig. 8. Outline of ACO for RPP of a mobile robot

TABLE II
ACO PARAMETER SPECIFICATIONS

ACO Properties	Parameter
Population of ants	Varies, depend on cases
Length of ants junction	Varies, depend on cases
Pheromone trail coefficient, β	5
Heuristic coefficient, α	5
Evaporation rate, ρ	0.5
Convergence condition	$C_{max} - C_{min} \leq 0.00001$

IV. EXPERIMENTAL RESULTS & DISCUSSION

A. Comparative study of GA and ACO in different complexity of free space nodes

Table III below shows the performances of GA and ACO. As shown, the time and number of iterations for both GA and ACO increases as the number of nodes increases (increase complexity). ACO seems to perform better. ACO has the robustness of the optimization algorithm compared to GA for this RPP purpose where it can still maintain the performances of time and number of iteration it takes to find optimal path in three different complexity of the environment as shown at Fig. 9 and Fig. 10.

TABLE II
PERFORMANCES OF GA AND ACO IN DIFFERENT COMPLEXITY OF ENVIRONMENT

Number of nodes	12 nodes		22 nodes		63 nodes	
Algorithm	GA	ACO	GA	ACO	GA	ACO
Time (sec)	72.12 44	9.337	242.7 088	32.566	2144 .84	814.8 19
Iteration	7.1	1.9	9.5	3.1	21.8	3.5

One of the main reasons for ACO's good performance is due to the behavior of the algorithm i.e. how it works to initialize the population. The way GA initialize the population is based on random approaches where the next node to be visited will determine easily by choosing any of random adjacent feasible node. With this random process, the algorithm need to go through the process of selection and some other process to choose the optimal node which will contribute to increment of time and iteration especially when the number of nodes is increasing. It was different with ACO where it has an efficient state transition rules which is efficient and can skip to the process of finding optimal path. With this efficient initialization approach, it will simultaneously help ACO to reduce the time and iteration while finding optimal path either in a simple environment or complex environment.

In addition, the quality of population in each generation will also influence algorithm performances. In ACO, the global and local updating approaches is efficient because it will improve the number of optimal path in each generation, this will make the solution faster with fewer number of iteration. However, this differs with GA where GA will carry the good population to produce the next child in the next generation however there is no guarantee to improve the solution rapidly as the child is produced in random approach. Thus the child may come from good categories or worst categories which will then influence the population of the next generation. Hence, this will affect the time and number of iterations that GA takes to find the solution will be greater than ACO's.

Furthermore, when the number of nodes increases, the size of chromosomes length also need to be adjusted based on the requirement in each case. Small number of nodes only needs a small number of lengths to allocate the paths within the chromosomes while the complex numbers of nodes will need a complex number of chromosomes. The increment of length will affect the whole process of GA to find optimal path because the population of the next child in each generation will be produced based on the cross over and mutation process. During this process, the point to be cross and mutate will be determined randomly based on the length of the chromosomes. The more the length, the more possibility of GA to have a variety of population which will cause the process of finding optimal path more challenging and simultaneously will contribute to the increment of time and iteration. It is different with ACO where the length of chromosomes size will not influence the next ant's population as it is based on heuristic and pheromone value carried by the previous ant's. Therefore, length will increase the time ants need to traverse from one node to another nodes and not influences the next node to be traversed by the ants.

With the increment of length usually GA also need to increase the population in order to get the optimal path. This will cause the process of GA to find path to become slower. In contrast with GA, for ACO, it is not necessary to increase the population because it will not affect the process. Thus this helps ACO to minimize the time and number of iterations.

Based on the reason and the results obtained above, ACO is practical to be use either in small or complex number of nodes compared to GA. This is because it can optimize the path in an efficient way compared to GA in ever different complexity of environment. ACO can find the optimal path and satisfy the optimization criteria at a faster rate than GA. In addition, the settings for ACO are compared to GA. In ACO, only the length needs to be changed in each cases, however for GA, when settings are changed, there is a need to ensure the balancing of the value of population, length and convergence criteria. From here it shows that the applications of ACO algorithm in RPP can be expanded by applying this algorithm in any type of global map that consists of multiple numbers of nodes, i.e. global map created in complex obstacles environment, global map created in a large indoor and outdoor environment etc. ACO can also be used to optimize not only global paths but also local paths. However, although GA is not as robust as ACO in optimizing the solution of RPP problem, it can still be used in applications that do not require fast solution. GA is able to provide solutions to path planning in complex environment cases at a slower rate.

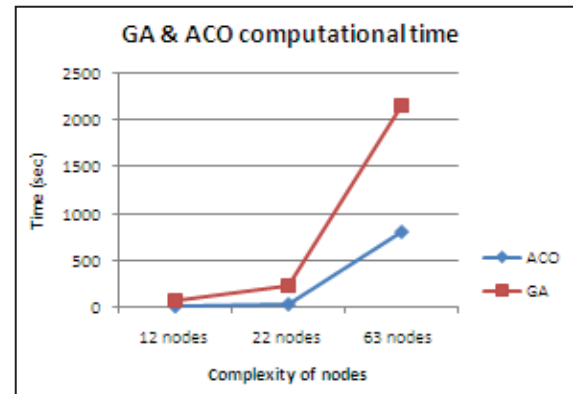


Fig. 9. GA & ACO computation time

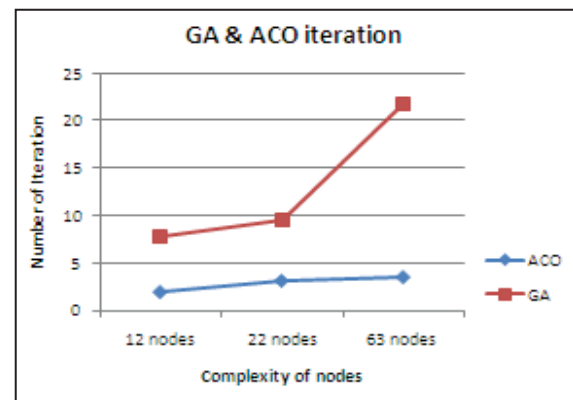


Fig. 10. GA & ACO iteration

V. CONCLUSION

GA and ACO were successfully implemented to find optimal path that satisfies the optimization criteria in the global static environment. Both algorithms performances were evaluated to determine the effectiveness of both algorithms when it is applied in different global static

environment. Findings from this research proved that ACO performance in terms of speed is much faster compared to GA while the number of iteration is less in ACO compared to GA in each different complexity of nodes in the global maps. In addition, the adjustment of ACO parameters to adapt with different complexities of nodes is also much easier compared to GA. Advantages and limitations of both algorithms can be further explored to expand the applications of both optimization algorithms in robot path planning research area.

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